

Physical Activity towards Health and Fitness II (Fitness Exercise)

CHAPTER III

STRENGTH and WEIGHT TRAINING

The hand is one of the most symbolized parts of the human body. It is often used to depict skill, action and purpose. Through our hands we may learn, create and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

Your body fat percentage will increase over time if you don't do anything to replace the lean muscle you lose over time. Strength training can help you preserve and enhance your muscle mass at any age.

Strength exercise, or resistance training, works your muscles by using resistance, like a dumbbell or your own body weight. This type of exercise increases lean muscle mass, which is particularly important for weight loss, because lean muscle burns more calories than other types of tissue. When people drop pounds, they can also lose muscle, so it's important to do resistance training to keep that muscle mass.

Strength training, also known as weight training or resistance training, involves the performance of physical exercises that are designed to improve strength and endurance.

Strength training specializes in the inducement of muscular contraction through the use of free weights, weight machines, and resistance bands which builds the strength, size of skeletal muscles, and anaerobic endurance. It strengthens and increases the amount of muscle mass in the body by making the muscles work harder than they're used to. It improves overall health and well-being because the bones, muscles, tendons and ligaments are strengthened and toughened. It also increases metabolism, improves cardiac function, and elevates good cholesterol. Strength training is most commonly seen as a weight-bearing activity. There are two terms concerning strength exercises: Repetitions and sets. A rep (repetition) is one full movement from starting point to finish. A set is a group of repetitions. The most basic training design is to have anywhere between 6 and 15 reps in a set, and performing 3 sets. Some people will go for 6 to 9 reps, some will go for 9-12 and some will go for 12-15 reps in a set. These numbers are for upper body strength training. The lower body reps must be anywhere between 15- 25 reps also with an approximate of three sets.

Strength training may also help you:

- **Develop strong bones.** By stressing your bones, strength training can increase bone density and reduce the risk of osteoporosis.
- **Manage your weight.** Strength training can help you manage or lose weight, and it can increase your metabolism to help you burn more calories.
- **Enhance your quality of life.** Strength training may enhance your quality of life and improve your ability to do everyday activities. Strength training can also

protect your joints from injury. Building muscle also can contribute to better balance and may reduce your risk of falls. This can help you maintain independence as you age.

- **Manage chronic conditions.** Strength training can reduce the signs and symptoms of many chronic conditions, such as arthritis, back pain, obesity, heart disease, depression and diabetes.
- **Sharpen your thinking skills.** Some research suggests that regular strength training and aerobic exercise may help improve thinking and learning skills for older adults.

Other benefits include:

- **Burn more fat:** Muscle is more metabolically active than fat, so the more you have, the more calories you burn all day.
- **Avoid injury:** Strong muscles mean you also have strong, supported bones and connective tissue. All of that contributes to a body that can withstand more stress than the bodies of people who don't do strength exercises.
- **Stay young and healthy:** Studies show that resistance training can enhance heart health, bone health, reduce blood pressure, lower cholesterol, increase bone density, reduce low back pain, improve sleep, and ease symptoms of arthritis and fibromyalgia.¹
- **Improve mood:** Research shows strength training can release feel-good endorphins to reduce anxiety and even fight depression.²
- **Boost confidence:** Anytime you master something, your confidence grows.

Some of the recommended activities are the following:

- **Body weight.** You can do many exercises with little or no equipment. Try pushups, pullups, planks, lunges and squats.
- **Resistance tubing.** Resistance tubing is inexpensive, lightweight tubing that provides resistance when stretched. You can choose from many types of resistance tubes in nearly any sporting goods store or online.
- **Free weights.** Barbells and dumbbells are classic strength training tools. If you don't have weights at home, you can use soup cans. Other options can include using medicine balls or kettle bells.
- **Weight machines.** Most fitness centers offer various resistance machines. You can invest in weight machines for use at home, too.
- **Cable suspension training.** Cable suspension training is another option to try. In cable suspension training, you suspend part of your body — such as your legs — while doing body weight training such as pushups or planks.

Know first!

1. Check with your doctor before beginning a strength training or aerobic fitness program.
2. consider warming up with brisk walking or another aerobic activity for five or 10 minutes. Cold muscles are more prone to injury than are warm muscles.
3. Choose a weight or resistance level heavy enough to tire your muscles after about 12 to 15 repetitions.

4. To give your muscles time to recover, rest one full day between exercising each specific muscle group.
5. If a strength training exercise causes pain, stop the exercise. Consider trying a lower weight or trying it again in a few days.
6. If you're new to strength training, work with a trainer or other fitness specialist to learn correct form and technique.

To get started

- Start with a short, simple program. Your goal is to do a routine that works for all muscle groups on two non-consecutive days a week. This will help you build a strong foundation and allow you to progress from week to week.
- Choose the right amount of weight to lift. The key is to use weights that are not too light and not too heavy. You'll know it's too light if you can do an entire set with minimal effort. It's too heavy if your form is sacrificed or it just feels too taxing. Just right is a challenging effort that you can do with proper form and control and without excess strain.
- Warm up first. Warm muscles are less susceptible to injury, so do 5 to 10 minutes of cardio or some warm-up sets of each exercise in your workout using a light, easy to lift weight.
- Focus on form. Good form means lets you reap all of the benefits of your workout and avoid injuries at the same time. To maintain proper form, pay attention to your posture (stand tall with chest lifted and abs held tight), move slowly (this ensures you're relying on muscles, not momentum, to do the lifting), and remember to breathe. Many people hold their breath while exerting, but exhaling during the hardest part of the exercise helps fuel the movement. Many modern interactive fitness mirrors bring the guidance of a trainer right into your living room.
- Give yourself at least a day of rest to recover. Rest days are crucial for building lean muscle tissue and preventing injury, so try not to work the same muscle groups two days in a row. Some people like to break up strength training by concentrating on their upper body one day and their lower body the next, and that's perfectly fine.
- Aim to challenge yourself, not overtax yourself. The first few weeks, focus on learning how to do each exercise rather than on how much weight you're lifting or how many exercises you're doing. You have plenty of time to build muscle.
- Change things up. After six or more weeks of consistent strength training, which is about the amount of time it takes to start seeing improvement in your body, you can change your routine to make it more difficult. Lifting the same weights for the same exercises every week will keep your body in the same place. You can modify weights or repetitions, choose different exercises, or change the order in which you do them. You only have to make one change at a time to make a difference, although more is often better.

Sample exercises for beginners:

- **Chest:** bench press, chest press, push-ups
- **Shoulders:** overhead press, lateral raise, front raise
- **Biceps:** biceps curls, hammer curls, concentration curls
- **Triceps:** triceps extensions, dips, kickbacks
- **Back:** one-arm row, back extensions, lat pulldowns
- **Abdominals:** crunches, reverse crunches, wood chops, pelvic tilts
- **Lower Body:** squats, lunges, leg press, deadlifts, calf raises

The Five Basic Strength Training Exercises

- A. **Push-ups**- Regular push-ups are prescribed to most people. However, for some who have health concerns, newbies and even females, they may start off with knee push-ups. Regular push-ups are performed with the body forming a straight line from the heels. The body is lowered until the chest is approximately one fist off the ground. The body is then pushed off the ground to return to its original position.
- B. **Superman**-From Push-ups prone lying position, lift both arms and legs. Try not to bend the legs; then lower arms and legs without relaxing the ground.
- C. **Crunches**- From the supine lying position while legs are bent and arms touching the ears, lift the head and shoulders with the shoulder blades lifting off completely from the ground. Return to original position while keeping the head elevated off the ground.
- D. **Side crunches**- Starting with legs together and bent, lie on one side. Extend the arm close to the ground 30-45 degrees from the body as support or balance. Position the other arm across the chest or have it float in front of it. Lift the shoulder closer to the ground while legs and the chest slightly face upward. Return to original position without letting the shoulder touch the ground.
- E. **Squats**-From an upright position with arms and hands crossed in front of chest or up, squat lower until the thighs are parallel to the ground. The chest and butt must be pushed out and the knee must not thrust forward beyond the toes. Return to original position, keeping both legs and slightly bent.